Meridian — Condensed Autopsy

A forensic measurement framework for RAG systems. The measurement layer is the contribution.

The thesis

RAG evaluation needs a deterministic trust anchor — a measurement layer that uses no LLM judges in its core computation, so that when a number moves, you know whether the pipeline changed or the measurement changed. Meridian's Layer 1 classifies every retrieved span against ground-truth character offsets using pure arithmetic. Layer 2 (LLM-judged correctness and faithfulness) sits on top, separate and labeled. The layers never contaminate each other.

The proof it works: four silent measurement bugs were caught by the framework's own verify-before-trust discipline before they could ship wrong numbers. Each was caught by systematic verification — not by the numbers looking wrong.

Headline numbers

Metric	Value	Type
Config-stack delta (correctness)	+7.9pp (60.5% → 68.3%)	Controlled
Config-stack delta (faithfulness)	+4.2pp (91.7% → 95.9%)	Controlled
Benchmark regime	4 corpora × 194 queries, combined index	—
Ruler calibration	3/4 corpora within ~2-3pp drift	Validated
Measurement bugs caught	4 (before shipping)	Methodology
Findings + corrections	47 findings, 7 corrections	Evidence trail

Architecture

Pipeline spine (top) observed by the measurement layer (bottom). Layer 1 observes retrieval output; Layer 2 observes answers. Arrows are one-way — measurement observes, never interferes.

The four caught bugs

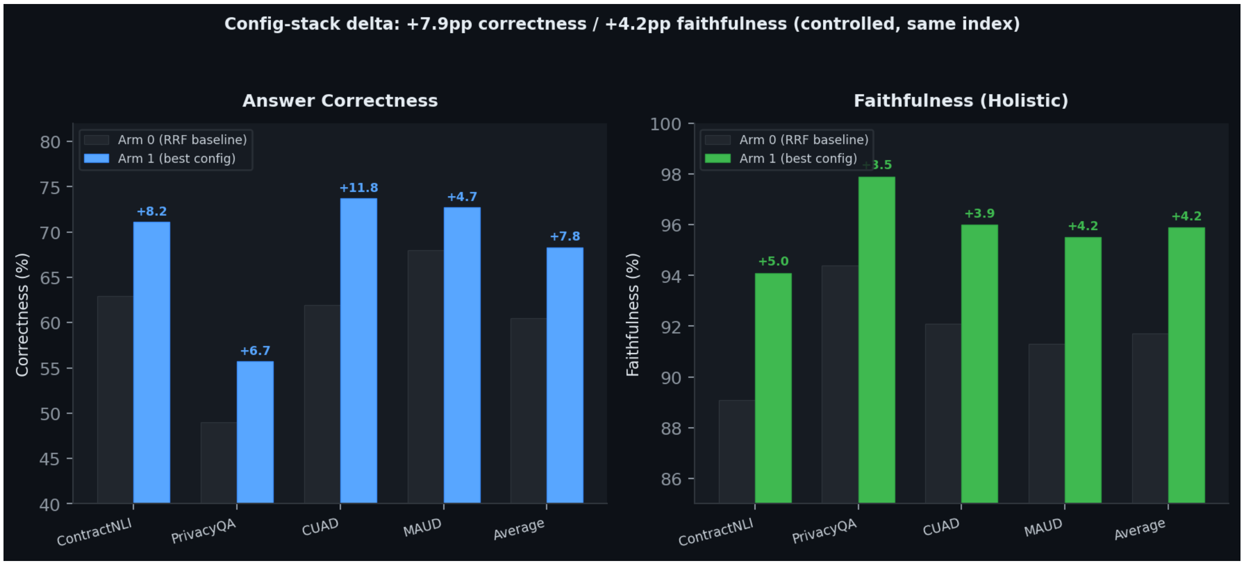
Each was caught by verification checks, not by the numbers looking wrong. A wrong number from a silent bug looks exactly like a correct number from a real measurement — the only defense is checking the instrument.

- **1. Pro model-ID alias (Finding 34).** PRO_MODEL="deepseek-chat" is a legacy DeepSeek alias that routes to flash, not Pro. Every prior "Pro" test was actually flash-vs-flash. Caught by billing check ($0.00 on Pro line) + API docs. Fixed: model ID corrected to "deepseek-v4-pro", served-model logging added to every synthesis call.
- **2. BM25 channel mismatch (Finding 45).** Combined-index dense retrieval searched 4,496 mini-doc chunks; BM25 searched the full 96,256-chunk CUAD parquet. Two channels measuring different pools, producing a hybrid that was neither combined nor per-corpus. Caught by crash on CUAD + post-mortem. Fixed: BM25 filtered to the same mini-doc set.
- **3. Zero-span offset bug (Finding 45).** Qdrant payloads don't store character spans (spans live in parquets). The combined-index builder defaulted to (0,0). All P@k/R@8 computed as near-zero — a plausible result, not an obvious crash. CUAD showed R@8=0.049 with 100% routing recall. Caught by sanity-check against calibration. Fixed: spans from parquets, recomputed. Real CUAD R@8=0.814.
- **4. Chunk-size inconsistency (Finding 46).** MAUD uses 2048-char chunks; the other three corpora use 512-char. Previously undocumented. Confounds MAUD's external P@k comparison with the paper's 500-char RCTS. Caught during external-baseline verification.

This is the project's thesis in practice: measurement rigor requires systematic verification at every step. The measurement layer's value is not just the taxonomy — it's the discipline of checking the instrument before trusting the reading.

Results

Config-stack delta (controlled)



Arm 0 (RRF baseline) vs Arm 1 (CC + routing + selector), same combined index. The delta isolates the method contribution.

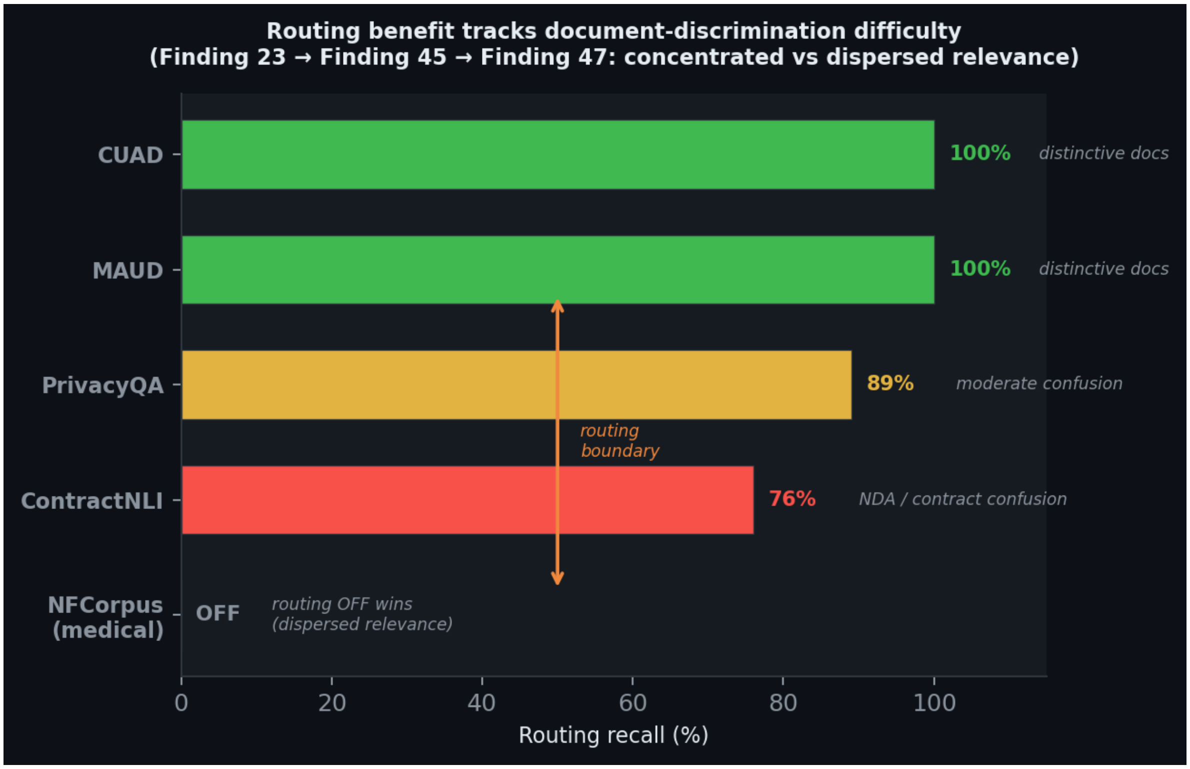
Corpus	Correct Arm 0→1	Faith Arm 0→1	P@1 Arm 1	R@8 Arm 1
ContractNLI	62.9→71.1 (+8.2)	89.1→94.1 (+5.0)	0.422	0.810
PrivacyQA	49.0→55.7 (+6.7)	94.4→97.9 (+3.5)	0.297	0.579
CUAD	61.9→73.7 (+11.8)	92.1→96.0 (+3.9)	0.394	0.814
MAUD	68.0→72.7 (+4.7)	91.3→95.5 (+4.2)	0.270	0.783
Average	60.5→68.3 (+7.9)	91.7→95.9 (+4.2)	—	—

External comparison (system-vs-system)

vs arXiv 2408.10343 Table 5 (RCTS, text-embedding-3-large, dense-only). 3 un-confounded corpora (512-char ≈ paper's 500-char). Full stack vs bare baseline — advantage bundles embedder + hybrid + fusion + routing. MAUD excluded (2048-char confound). ContractNLI caveated (benchmark provenance).

Corpus	Meridian P@1	RCTS P@1	Meridian R@8	RCTS R@8
ContractNLI	0.422	0.066	0.810	0.250
CUAD	0.394	0.020	0.814	0.317
PrivacyQA	0.297	0.144	0.579	0.424

Routing boundary



Routing benefit tracks document-discrimination difficulty. 100% on distinctive corpora (CUAD, MAUD), 76% on homogeneous (ContractNLI), OFF on dispersed-relevance medical text (NFCorpus).

NFCorpus transfer (retrieval stack only)

nDCG@10 = 0.399, above classic BEIR baselines (BM25 0.325, BM25+CE 0.350, contriever 0.328). CC fusion transfers (+5.6pp over RRF). The span-forensic framework was NOT exercised (BEIR has no character spans). Routing correctly self-disabled. The genuine finding: routing is a concentrated-relevance technique — the sweep auto-detects when it helps vs hurts.

Limitations

- **Routing degrades to 76%** on ContractNLI in the combined regime (NDA/commercial-contract confusion). 47/194 queries get zero correct-document chunks.
- **Routing hurts on dispersed relevance.** Confirmed on NFCorpus. Routing is concentrated-relevance only.
- **Span framework requires character-span ground truth.** Does not apply to document-level benchmarks (most of BEIR). This is a scope boundary.
- **MAUD external comparison confounded** by 2048-char vs 500-char chunk granularity.
- **Affirmative-only evaluation.** Correctness measures recall, not false-positive rate.
- **Synthesis variance band ±2-4pp.** Deltas below this are indistinguishable from nondeterminism.
- **External multipliers are system-level,** not method-level. The advantage bundles embedder + hybrid + fusion + routing.

What Meridian is not

Not an autonomous research agent (abandoned in v1). Not a SOTA-everywhere claim (beats classic baselines, not modern dense SOTA). Not a framework-transfer claim (span taxonomy ran on LegalBench-RAG only; NFCorpus was retrieval-transfer). Not a tutorial.

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47 findings with 7 corrections — the full evidence trail is in docs/DECISIONS.md. The complete methodology and results are in docs/REPORT.md.